K_0 OF THE BOUNDED DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Long exact cohomology sequence and canonical delta functor. equation-long-exact-cohomology-sequence-D, section-canonical-delta-functor (same chapter, cited in-span; not carded). Give the distinguished triangle attached to a short exact sequence of complexes and its boundary maps.
- Truncation sections. homology-section-truncations (cross-chapter, cited in-span; not carded). Feeds the additivity chase for the Euler characteristic map.

Source reference. Original equation 05ST: sources/stacks-derived/pinned-original-source.tex, source lines 3857–3857. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0118: focused-seeds/geometry/S9-der-K-bounded-derived/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 014Z: sources/stacks-derived/pinned-original-source.tex, source lines 4118–4118. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Then there is a canonical identification $K_0(D^b(\mathcal{A})) = K_0(\mathcal{A})$ of zeroth K-groups.*

Proof. Given an object A of $\mathcal{A}$ denote $A[0]$ the object A viewed as a complex sitting in degree 0. If $0 \to A \to A' \to A'' \to 0$ is a short exact sequence, then we get a distinguished triangle $A[0] \to A'[0] \to A''[0] \to A[1]$, see Section 014Z. This shows that we obtain a map $K_0(\mathcal{A}) \to K_0(D^b(\mathcal{A}))$ by sending $[A]$ to $[A[0]]$ with apologies for the horrendous notation.

On the other hand, given an object X of $D^b(\mathcal{A})$ we can consider the element

$$c(X) = \sum (-1)^i [H^i(X)] \in K_0(\mathcal{A})$$

Given a distinguished triangle $X \to Y \to Z$ the long exact sequence of cohomology (05ST) and the relations in $K_0(\mathcal{A})$ show that $c(Y) = c(X) + c(Z)$. Thus c factors through a map $c : K_0(D^b(\mathcal{A})) \to K_0(\mathcal{A})$.

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We want to show that the two maps above are mutually inverse. It is clear that the composition $K_0(\mathcal{A}) \to K_0(D^b(\mathcal{A})) \to K_0(\mathcal{A})$ is the identity. Suppose that $X^\bullet$ is a bounded complex of $\mathcal{A}$. The existence of the distinguished triangles of "stupid truncations" (see Homology, Section 0118)

$$\sigma_{\geq n}X^\bullet \to \sigma_{\geq n-1}X^\bullet \to X^{n-1}[-n+1] \to (\sigma_{\geq n}X^\bullet)[1]$$

and induction show that

$$[X^\bullet] = \sum (-1)^i [X^i[0]]$$

in $K_0(D^b(\mathcal{A}))$ (with again apologies for the notation). It follows that the composition $K_0(\mathcal{A}) \to K_0(D^b(\mathcal{A}))$ is surjective which finishes the proof. $\square$